

		$\omega = 5.3 \text{ rad s}^{-1}$
12	A	loss in $K.E.$ = gain in $G.P.E.$ $\frac{1}{2}mv_{esc}^2 - 0 = U_{\infty} - U_{surface} \quad (\text{assuming that } K.E. \text{ at infinity is zero})$ $\frac{1}{2}mv_{esc}^2 = \left[0 - \left(-\frac{GMm}{R} \right) \right] \quad \text{since } U_{\infty} = 0.$ $v_{esc} = \sqrt{\frac{2GM}{R}}$
13	C	$\sqrt{\frac{2}{\rho^2}} \sqrt{3RT}$